

HILLSBOROUGH, United Kingdom

WMO: 039200

Lat: 54.485N

Lon: 6.097W

Elev: 125

StdP: 14.63

Time Zone: 0.00 (EUW)

Period: 94-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.9	28.0	21.6	16.3	30.1	24.4	18.6	32.3	28.4	45.9	25.2	46.0	1.5	230	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.4	73.0	63.2	69.9	61.8	67.3	60.3	64.8	70.6	63.1	68.0	61.6	65.9	8.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	85.1	67.4	60.9	80.2	65.7	59.5	76.3	64.0	29.9	71.0	28.6	68.1	27.5	65.8	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	20.1	17.8	18.4	76.2	5.2	3.0	14.7	78.3	11.6	80.1	8.7	81.8	4.9	84.0		
(5)			17.9	66.6	5.2	2.2	14.2	68.2	11.1	69.5	8.2	70.7	4.4	72.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	40.9	41.5	43.5	46.1	51.1	55.2	58.9	59.2	55.0	50.2	45.4	40.8		
	DBStd	8.22	5.30	5.22	4.85	4.90	4.78	3.98	3.69	4.20	3.93	5.21	5.35	6.26		
	HDD50	1433	285	243	207	136	45	5	0	0	5	63	155	289		
	HDD65	5843	748	659	666	568	431	295	191	187	300	458	589	751		
	CDD50	1077	2	4	5	18	79	160	277	286	156	70	16	4		
	CDD65	13	0	0	0	0	0	1	3	8	1	0	0	0		
	CDH74	55	0	0	0	0	0	11	14	30	0	0	0	0		
	CDH80	2	0	0	0	0	0	1	0	1	0	0	0	0		
(14) Wind	WSAvg	7.9	8.9	10.2	8.7	8.1	7.4	7.5	6.8	6.2	6.4	8.4	7.9	8.1		
(15) Precipitation	PrecAvg	40.2	3.9	2.8	2.8	2.8	3.0	2.8	3.0	3.7	4.1	3.9	4.0	3.3		
	PrecMax	49.6	6.1	6.9	4.5	5.2	4.9	4.6	7.3	6.0	7.0	6.7	6.2	6.3		
	PrecMin	29.8	1.7	0.6	1.5	1.3	0.9	1.3	1.5	0.6	0.5	1.8	2.2	0.6		
	PrecStd	5.2	1.2	1.5	1.0	1.1	1.2	1.0	1.4	1.5	1.8	1.7	1.0	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	56.3	57.1	64.2	70.2	75.1	76.0	77.8	70.0	63.9	59.2	57.1		
		MCWB	52.2	52.9	51.0	54.2	59.3	62.9	65.6	66.5	63.7	59.1	56.7	54.9		
	2%	DB	52.8	53.8	54.9	59.8	66.3	68.8	72.4	73.5	66.6	61.4	56.9	54.0		
		MCWB	50.0	50.5	50.2	53.0	57.5	59.5	62.9	64.6	60.9	57.7	54.5	51.5		
	5%	DB	50.9	51.8	53.4	56.5	62.9	65.3	69.4	70.0	64.4	59.8	55.0	52.1		
		MCWB	48.4	48.9	49.2	50.9	55.8	57.5	61.5	62.7	58.7	56.5	52.5	49.6		
	10%	DB	49.2	49.8	51.6	54.0	59.6	62.7	66.7	67.1	62.4	58.3	53.4	49.5		
		MCWB	47.0	46.8	48.0	49.2	53.4	56.0	60.0	60.8	57.2	55.2	51.2	47.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	53.1	53.3	56.6	60.5	63.7	67.4	67.9	64.7	60.0	56.8	55.2		
		MCDB	54.5	55.7	55.2	61.1	67.7	72.5	75.0	74.5	68.4	63.2	58.4	56.8		
	2%	WB	50.4	51.0	51.5	53.8	58.2	60.5	64.4	65.4	62.3	58.4	54.9	51.8		
		MCDB	52.6	53.4	54.1	58.6	64.6	66.3	69.8	71.9	65.4	61.0	56.4	53.7		
	5%	WB	48.6	49.3	50.0	51.8	56.1	58.9	62.5	63.6	60.0	57.2	53.1	49.9		
		MCDB	50.5	51.6	52.7	55.8	61.6	64.2	67.4	68.6	63.0	59.5	54.5	52.0		
	10%	WB	47.2	47.0	48.4	49.9	54.2	57.2	60.9	62.0	58.2	55.6	51.5	47.4		
		MCDB	49.0	49.4	51.1	53.4	58.7	61.7	65.6	66.2	61.2	57.9	53.3	49.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	10.5	11.7	13.5	14.4	13.0	13.2	14.4	13.7	11.3	10.1	9.4		
		MCDBR	9.5	10.8	13.1	15.5	18.9	17.4	18.0	18.0	16.1	11.9	9.2	10.6		
	5% WB	MCWBR	8.8	9.9	9.4	9.9	11.8	10.2	10.3	10.1	10.5	9.3	8.3	9.9		
		MCWBR	9.5	10.2	11.1	13.7	17.2	15.8	15.7	15.1	13.8	11.0	9.1	10.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.333	0.344	0.369	0.383	0.388	0.381	0.382	0.366	0.359	0.344	0.326	0.321		
		taud	2.240	2.318	2.286	2.281	2.312	2.365	2.365	2.425	2.438	2.419	2.307	2.207		
	Ebn at Noon	Ebn at Noon	187	233	253	267	272	275	272	270	256	231	190	161		
		Edh at Noon	20	25	32	37	38	36	36	32	28	23	18	16		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	185	400	746	1200	1523	1547	1440	1182	847	478	231	137		
	RadStd	RadStd	20	37	69	109	134	154	123	94	47	46	25	13		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page